

## 6.6 Tutte's flow conjectures

How can we determine the flow number of a graph? Indeed, does every (bridgeless) graph have a flow number, a  $k$ -flow for some  $k$ ? Can flow numbers, like chromatic numbers, become arbitrarily large? Can we characterize the graphs admitting a  $k$ -flow, for given  $k$ ?

Of these four questions, we shall answer the second and third in this section: we prove that every bridgeless graph has a 6-flow. In particular, a graph has a flow number if and only if it has no bridge. The question asking for a characterization of the graphs with a  $k$ -flow remains interesting for  $k = 3, 4, 5$ . Partial answers are suggested by the following three conjectures of Tutte, who initiated algebraic flow theory on graphs.

The oldest and best known of the Tutte conjectures is his *5-flow conjecture*:

**Five-Flow Conjecture.** (Tutte 1954)

*Every bridgeless multigraph has a 5-flow.*

Which graphs have a 4-flow? By Proposition 6.4.4, the 4-edge-connected graphs are among them. The Petersen graph (Fig. 6.6.1), on the other hand, is an example of a bridgeless graph without a 4-flow: since it is cubic but not 3-edge-colourable, it cannot have a 4-flow by Proposition 6.4.5 (ii).

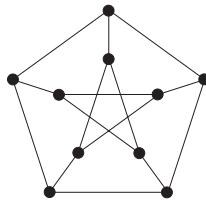


Fig. 6.6.1. The Petersen graph

Tutte's *4-flow conjecture* states that the Petersen graph must be present in every graph without a 4-flow:

**Four-Flow Conjecture.** (Tutte 1966)

*Every bridgeless multigraph not containing the Petersen graph as a minor has a 4-flow.*

By Proposition 1.7.3, we may replace the word ‘minor’ in the 4-flow conjecture by ‘topological minor’.

Even if true, the 4-flow conjecture will not be best possible: a  $K^{11}$ , for example, contains the Petersen graph as a minor but has a 4-flow, even a 2-flow. The conjecture appears more natural for sparser graphs; a proof for cubic graphs was announced in 1998 by Robertson, Sanders, Seymour and Thomas, which however still has not been published in full.

*snark* A cubic bridgeless graph or multigraph without a 4-flow (equivalently, without a 3-edge-colouring) is called a *snark*. The 4-flow conjecture for cubic graphs says that every snark contains the Petersen graph as a minor; in this sense, the Petersen graph has thus been shown to be the smallest snark. Snarks form the hard core both of the four colour theorem and of the 5-flow conjecture: the four colour theorem is equivalent to the assertion that no snark is planar (exercise), and it is not difficult to reduce the 5-flow conjecture to the case of snarks.<sup>6</sup> However, although the snarks form a very special class of graphs, none of the problems mentioned seems to become much easier by this reduction.<sup>7</sup>

### Three-Flow Conjecture. (Tutte 1972)

*Every multigraph without a cut consisting of exactly one or exactly three edges has a 3-flow.*

Again, the 3-flow conjecture will not be best possible: it is easy to construct graphs with three-edge cuts that have a 3-flow (exercise).

By our duality theorem (6.5.3), all three flow conjectures are true for planar graphs and thus motivated: the 3-flow conjecture translates to Grötzsch's theorem (5.1.3), the 4-flow conjecture to the four colour theorem (since the Petersen graph is not planar, it is not a minor of a planar graph), the 5-flow conjecture to the five colour theorem.

We finish this section with the main result of the chapter:

### Theorem 6.6.1. (Seymour 1981)

*Every bridgeless multigraph has a 6-flow.*

*(3.3.6)* *Proof.* We prove by induction on  $|G|$  that, given any bridgeless multi-  
*(6.1.3)* graph  $G = (V, E)$  and a vertex  $u$  of  $G$ , there exists a  $(\mathbb{Z}_2 \times \mathbb{Z}_3)$ -flow  
*(6.3.2)*  $f = f_2 \times f_3$  on  $G$  such that  $f_2$  is zero on all the edges at  $u$ . By Corol-  
*(6.3.3)* lary 6.3.2 and Theorem 6.3.3, then,  $G$  will also have a 6-flow.  
*u*

The induction starts trivially with  $|G| = 1$ , since loops are ignored in (F1) and (F2). So let  $G$  be given for the induction step. This is trivial if  $G$  is disconnected or has a cutvertex, so we assume that  $G$  is 2-connected. Then  $G - u$  is connected.

<sup>6</sup> The same applies to another well-known conjecture, the *cycle double cover conjecture*; see Exercise 17.

<sup>7</sup> That snarks are elusive has been known to mathematicians for some time; cf. Lewis Carroll, *The Hunting of the Snark*, Macmillan 1876.

If  $G - u$  has a bridge,  $e$  say, let  $\{V_1, V_2\}$  be the partition of  $V(G - u)$  underlying its cut  $\{e\}$ . For  $i = 1, 2$  the sets  $U_i := V_i \cup \{u\}$  are connected in  $G$ ; we contract them to form  $G_i := G/U_i$  with contracted vertex  $u_i$ . By the induction hypothesis,  $G_i$  has a  $(\mathbb{Z}_2 \times \mathbb{Z}_3)$ -flow  $f^i = f_2^i \times f_3^i$  in which  $f_2^i$  is zero on  $u_i$ . Replacing  $f_3^1$  with  $-f_3^1$  if necessary we may assume that it agrees with  $f_3^2$  on  $\vec{e}$  and  $\bar{e}$ . Now  $f := f^1 \cup f^2$  is a  $(\mathbb{Z}_2 \times \mathbb{Z}_3)$ -flow on  $G$  whose first component  $f_2$  is zero on all the edges at  $u$ , as desired.

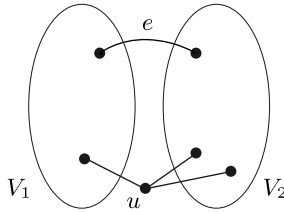


Fig. 6.6.2. Constructing  $f$  when  $G - u$  has a bridge

It remains to consider the case that  $G - u$  is bridgeless. As  $G - u$  is connected, this makes it 2-edge-connected. Let  $H$  be the union of two edge-disjoint paths in  $G - u$  between distinct neighbours of  $u$ ; these paths exist by Menger's theorem 3.3.6 (ii). Note that  $H$  is an even connected submultigraph of  $G - u$  (Figure 6.6.3).

$H$

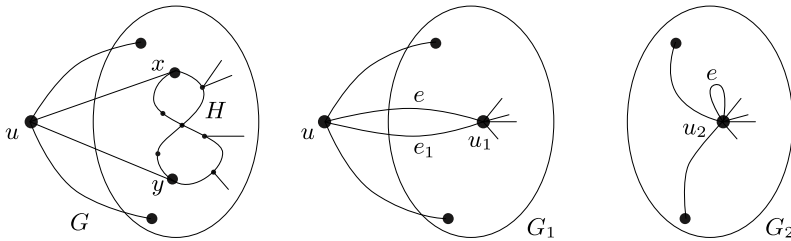


Fig. 6.6.3. Constructing  $f$  when  $G - u$  has no bridge

Let  $G_1$  be the minor of  $G$  obtained by contracting the edges of  $H$  to a vertex  $u_1$ , and let  $G_2$  be obtained from  $G_1$  by contracting a  $u-u_1$  edge  $e_1$  to a vertex  $u_2$ . Both  $G_1$  and  $G_2$  are bridgeless, because contracting edges never creates a bridge. By the induction hypothesis,  $G_2$  has a  $(\mathbb{Z}_2 \times \mathbb{Z}_3)$ -flow  $f_2^2 \times f_3^2$  with  $f_2^2$  zero on all the edges at  $u_2$ . By Lemma 6.1.3,  $f_3^2$  extends to a  $\mathbb{Z}_3$ -circulation  $f_3^1$  on  $G_1$  that is non-zero everywhere except possibly on  $e_1$ . As  $G_1$  has at least two  $u-u_1$  edges, however, we can modify  $f_3^1$  to make it non-zero on all the edges of  $G_1$ , including  $e_1$ . Applying Lemma 6.1.3 once more, we extend  $f_3^1$  to a  $\mathbb{Z}_3$ -circulation  $f_3$  on  $G$  that is non-zero on all edges not in  $H$ .

$G_1, u_1$

$e_1$

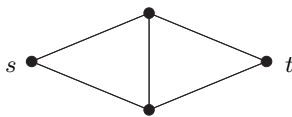
$G_2, u_2$

As  $H$  is even and  $f_2^2$  maps all the edges at  $u_2$  to zero, we can extend  $f_2^2$  to a  $\mathbb{Z}_2$ -circulation  $f_2$  on  $G - e_1$  that maps all its edges at  $u$  to

zero, by letting  $f_2(\vec{e}) := \bar{1} \in \mathbb{Z}_2$  for all edges  $e$  of  $H$ . Setting  $f_2$  to zero on  $e_1$  too then turns it into a  $\mathbb{Z}_2$ -circulation on  $G$ . Now  $f := f_2 \times f_3$  is a  $(\mathbb{Z}_2 \times \mathbb{Z}_3)$ -flow on  $G$  with  $f_2$  zero on all the edges at  $u$ , as desired.  $\square$

## Exercises

1.  $^-$  Prove Proposition 6.2.1 by induction on  $|S|$ .
2. (i)  $^-$  Given  $n \in \mathbb{N}$ , find a capacity function for the network below such that the algorithm from the proof of the max-flow min-cut theorem will need more than  $n$  augmenting paths  $W$  if these are badly chosen.



- (ii)  $^+$  Show that, if all augmenting paths are chosen as short as possible, their number is bounded by a function of the size of the network.
3.  $^+$  Derive Menger's Theorem 3.3.5 from the max-flow min-cut theorem.  
(Hint. The edge version is easy. For the vertex version, apply the edge version to a suitable auxiliary graph.)
4.  $^-$  Let  $f: \vec{E} \rightarrow H$  be a circulation on  $G$  and  $g: H \rightarrow H'$  a group homomorphism. Show that  $g \circ f$  is a circulation on  $G$ . Is  $g \circ f$  an  $H'$ -flow if  $f$  is an  $H$ -flow?
5. View the group of circulations on a graph with values in  $\mathbb{Z}_2$  as a vector space over  $\mathbb{Z}_2$ . Find a space in Chapter 1.9 to which it is isomorphic, and write down an explicit isomorphism.
6. Let  $H$  be an abelian group,  $G = (V, E)$  a connected graph,  $T$  a spanning tree, and  $f$  a map from the orientations of the edges in  $E \setminus E(T)$  to  $H$  that satisfies (F1). Show that  $f$  extends uniquely to a circulation on  $G$  with values in  $H$ .
7. (continued)

Let  $\mathcal{V}_H = \mathcal{V}_H(G)$  be the group of all maps  $V \rightarrow H$ , and  $\mathcal{E}_H = \mathcal{E}_H(G)$  the group of all maps  $\vec{E} \rightarrow H$  satisfying (F1), both with pointwise addition. Every  $\varphi \in \mathcal{V}_H$  defines a  $\psi \in \mathcal{E}_H$  by  $\psi(e, x, y) := \varphi(y) - \varphi(x)$ .

- (i) Show that these  $\psi$  form a subgroup  $\mathcal{B}_H = \mathcal{B}_H(G)$  of  $\mathcal{E}_H$  with  $\mathcal{B}_H = \{ \psi \in \mathcal{E}_H \mid \psi(\vec{C}) = 0 \text{ for every oriented cycle } C \subseteq G \}$ , where  $\psi(\vec{C}) := \sum_{\vec{e} \in \vec{C}} \psi(\vec{e})$ .
- (ii) Show that every map  $\vec{E}(T) \rightarrow H$  satisfying (F1) extends uniquely to a map in  $\mathcal{B}_H$ .